

12 (**long**)

(a) Consider the gradient function

$$\nabla f = (ye^x + x^2 + 1) \mathbf{i} + \left( \frac{1}{(y+1)(y+2)} + e^x \right) \mathbf{j}$$

(i) Find the function  $f(x, y)$ . [9]

(ii) Calculate  $\frac{\partial^2 f}{\partial y \partial x}$  and  $\frac{\partial^2 f}{\partial x \partial y}$ . [3]

(iii) Determine the gradient of the function  $f(x, y)$  at the origin along the curve  $\mathbf{c} = (e^{3u} - 1) \mathbf{i} + u \mathbf{j}$  in the direction of increasing  $u$ . [8]

(b) A surface is defined by the equation

$$\mathbf{r} = u^2 v \mathbf{i} + v^2 \mathbf{j} + uv \mathbf{k}$$

Find the normal to the surface at the point where  $u = 1$  and  $v = 2$ . [10]

**END OF PAPER**